

Perspective

Antarctica and digital twins: towards integrated outreach, open science and research station management

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Abstract

Antarctica constitutes a natural laboratory dedicated exclusively to peaceful and scientific purposes. In this context, Antarctic research stations represent the operational hubs for scientific activity and the primary sites of human presence on the continent. However, their remoteness, logistical complexity and strong seasonality constrain public visibility, accessibility and the monitoring of scientific operations. This perspective paper explores the potential of spatial digital twins (SDTs) as integrative tools to address these limitations and proposing a conceptual framework for their application in Antarctic research stations. This framework leverages virtual platforms for the operational and scientific monitoring of infrastructures, as well as for open data dissemination, decision support and improved accessibility, in line with the principles of the Antarctic Treaty. By synthesizing spatial data, sensor networks and real-time streams, SDTs enable the continuous monitoring of complex socio-technical systems. Drawing on their transformative role in fields such as urban planning and industrial management, SDTs can enhance operational management, scientific tracking and open data dissemination. Finally, the discussion addresses the potential technical, logistical and institutional challenges associated with their implementation into Antarctic research stations, such as data availability, cost-benefit balancing, standardization, scalability for large-scale stations and long-term maintenance. Despite these hurdles, SDTs have the potential to become a valuable and scalable standard for monitoring, managing and communicating scientific activity in one of the most remote and scientifically significant regions on the planet.

Keywords: Antarctic research; Geographic Information Systems (GIS); open research; polar regions; science communication; spatial data

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Introduction

Antarctica, including the Southern Ocean and its surrounding islands, constitutes a territory devoted exclusively to peaceful and scientific purposes, as established by the Antarctic Treaty (The Secretariat of the Antarctic Treaty 1991). This historical and legislative framework promotes international cooperation, open science and data exchange as fundamental principles for advancing knowledge in one of the most pristine, remote and sensitive systems on the planet (Dudeney & Walton 2012, Dodds 2024b, Leary & Jabour 2024). Scientific activity in the region is primarily conducted through camps and research stations distributed across the continent, from the geographical South Pole to coastal and island environments, with a notable concentration in the Antarctic Peninsula (COMNAP 2017). Since the mid-twentieth century, these stations and their associated infrastructure have played a key role in sustaining the human presence in Antarctica; evolving from temporary settlements and small clusters of structures into complex scientific platforms, with a progressive increase in both

their number and operational capacity (Nielsen 2013, Hudson 2023, Menéndez-Blázquez 2023). Today, Antarctic research stations function as platforms for environmental observation, scientific laboratories and communication and logistical hubs, accommodating diverse personnel such as researchers, logistical staff (sometimes supported by the military) and other stakeholders that enable research activities at these latitudes. However, their geographical isolation and generally limited human presence pose significant challenges for the systematic monitoring of activities and for the transparent dissemination of Antarctic science to the international community and society at large.

As highlighted by Nighat Amin (International Polar Foundation), ‘An Antarctic station is not just a building, but rather it has become a process by which we can redefine the world in which we live, a means to challenge existing realities and create new ones’ (Nielsen 2013). This vision reframes research stations not merely as static infrastructures but as dynamic socio-technical systems in which scientific, logistical, environmental and human dimensions continuously interact. Managing this complexity demands a novel technological approach capable of mirroring such dynamic interactions. This perspective paper explores and illustrates a conceptual framework for the application of digital twins as integrative platforms for these research stations. The novelty of this contribution lies in bridging the gap between advanced digital

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modelling and the unique constraints of remote polar logistics, offering a roadmap to harmonize fragmented data sources into a coherent digital ecosystem. In this context, digital twins have emerged as key technologies for representing complex systems through virtual models that are continuously updated with real-time data (Crespi *et al.* 2023). Their ability to integrate diverse data sources, off-site sensors, predictive models, interactive dashboards and advanced visualizations (Yan *et al.* 2019, Botín-Sanabria *et al.* 2022, Fahim *et al.* 2022) makes them particularly well suited for remote, dynamic and multidisciplinary environments such as Antarctica: the stations comprise critical infrastructures and heterogeneous data streams, rendering each station an autonomous system well suited to digital twin-based modelling. Furthermore, this perspective paper addresses the technical and logistical challenges involved, contending that these systems could strengthen scientific, management and outreach activities in line with the principles of transparency and international cooperation underpinning the Antarctic Treaty System.

Spatial digital twins

Digital twins are dynamic computational models that replicate real-world systems, physical environments and their interactions with other elements, continuously integrating observational data to reflect their state and evolution in near-real time (Crespi *et al.* 2023). When these models are built upon a Geographic Information System (GIS) framework, they are referred to as 'GIS-based', 'geospatial' or 'spatial digital twins' (hereinafter SDTs; Lehner & Dorffner 2020), enabling the visualization and analysis of spatial information across multiple dimensions (2D/3D). SDTs allow the simulation and prediction of past or future scenarios, as well as the assessment of management alternatives, making them essential tools for operational and strategic decision-making (Crespi *et al.* 2023). These technologies are capable of adapting to diverse requirements, and they have been successfully applied in industry, energy system monitoring and optimization, agriculture, environmental conservation and smart cities (Schrotter & Hürzeler 2020, Bayer & Pruckner 2023, Clementi *et al.* 2024, Durden 2025, Melesse 2025). Although these foundational principles are today becoming well-established, they have yet to be implemented in the scientific context of research stations in the Antarctic region, representing a unique opportunity for innovation for both SDTs themselves and polar science.

Digital twins in Antarctic research stations

Translating this technology to Antarctic research stations requires a tailored architecture that addresses their structural and operational diversity, as well as the connectivity provided by research and logistical vessels, which act as vital mobile nodes within the broader Antarctic network (Fig. 1). In practice, this system is envisioned as a web-based geospatial platform that serves as the integrative hub for the entire digital ecosystem (e.g. ArcGIS Enterprise, TerriaJS, etc.). Far from being a single piece of software, the SDT architecture is composed of different modular components: 1) a data-integration and storage layer for harvesting real-time sensor feeds and heterogeneous datasets (e.g. using FIWARE or Apache Kafka platforms for ingestion and PostgreSQL/PostGIS for data management), 2) a GIS-based interface for spatial data management (as mentioned previously), 3) modular dashboards for real-time monitoring and data visualization and 4) interactive 3D and Building Information Modelling (BIM) modules for

infrastructure representation (e.g. ESRI ecosystem, CesiumJS, Autodesk Platform Services, Geolibre.app, etc.; Figs 2 & 3). Furthermore, this web-based environment functions as a centralized knowledge repository, enabling direct integration of external project links, real-time research documentation and multimedia visualization tools, among other benefits. This modular framework addresses the aims of the SDTs, serving as a dynamic interface for the seamless integration of energy systems, logistical flows and scientific datasets, providing a unified environment for both internal station management and public outreach.

Science outreach and dissemination

The implementation of SDTs in an Antarctic research station should be rooted in the fundamental pillar of scientific outreach, enabling both the general public and the scientific community to access real-time information regarding ongoing activities. A SDT can serve as a valuable outreach hub by integrating the basic geographical context with detailed data from diverse sources (Fig. 2), effectively bridging the gap between site location, ongoing research projects and related scientific findings for a broader audience. For example, the yearly Spanish Army penguin sponsor campaign (https://ejercito.defensa.gob.es/en/unidades/Antartica/antartica/apadrinamiento/index.html?__locale=en) aims to raise awareness regarding the conservation of these sentinel species on Deception Island (South Shetlands Islands, Antarctic Peninsula). However, apart from mentioning the name of the island, its website does not provide additional information regarding its location, nor the species breeding areas, conservation status, biology, latest scientific findings for the species, and so on.

To provide insight into stations' human presence or occupancy models, SDTs could consolidate dashboards or similar tools summarizing the composition and diversity of station personnel (e.g. number of researchers, logistical and technical staff, as well as their nationalities; Fig. 2), reflecting the collaborative nature of Antarctic science (Karplus *et al.* 2022). Furthermore, they could integrate information on ongoing research projects (including disciplines, research teams and affiliated institutions), thereby facilitating the dissemination of ongoing campaign objectives and strengthening networking within the scientific community (Fig. 2; Jang *et al.* 2020, Vila *et al.* 2025). In addition, some of these ongoing research projects rely on the use of remote and automated logger devices that collect publicly accessible data (e.g. open-source meteorological sensors, fixed cameras and renewable energy data; as example is the Spanish Antarctic research station BAE Juan Carlos I in Livingston Island; cndp.utm.csic.es/rt/), which can be seamlessly integrated within a SDT (Fig. 2).

Many Antarctic stations and research facilities are named after pioneering scientists and explorers, yet their contributions often remain unknown to the public. SDTs provide a unique opportunity to integrate historical and biographical contexts, providing recognition for their figures (e.g. the late Josefina Castellví, a pioneering figure in Antarctic science and the first woman to serve as director of the Spanish Antarctic research station; Fig. 3a), thereby connecting scientific outputs with the people behind them.

SDTs further enhance outreach by enabling dynamic 3D visualizations of research station infrastructure and the surrounding landscape (Yan *et al.* 2019, Lehner & Dorffner 2020). These realistic and immersive visualizations can be reconstructed from BIM and photogrammetric data (using drone imagery or on-site cameras; e.g. photorealistic techniques such as orthomosaics or Gaussian splat cloud-points; Román *et al.* 2024, Yan *et al.* 2025), and they can

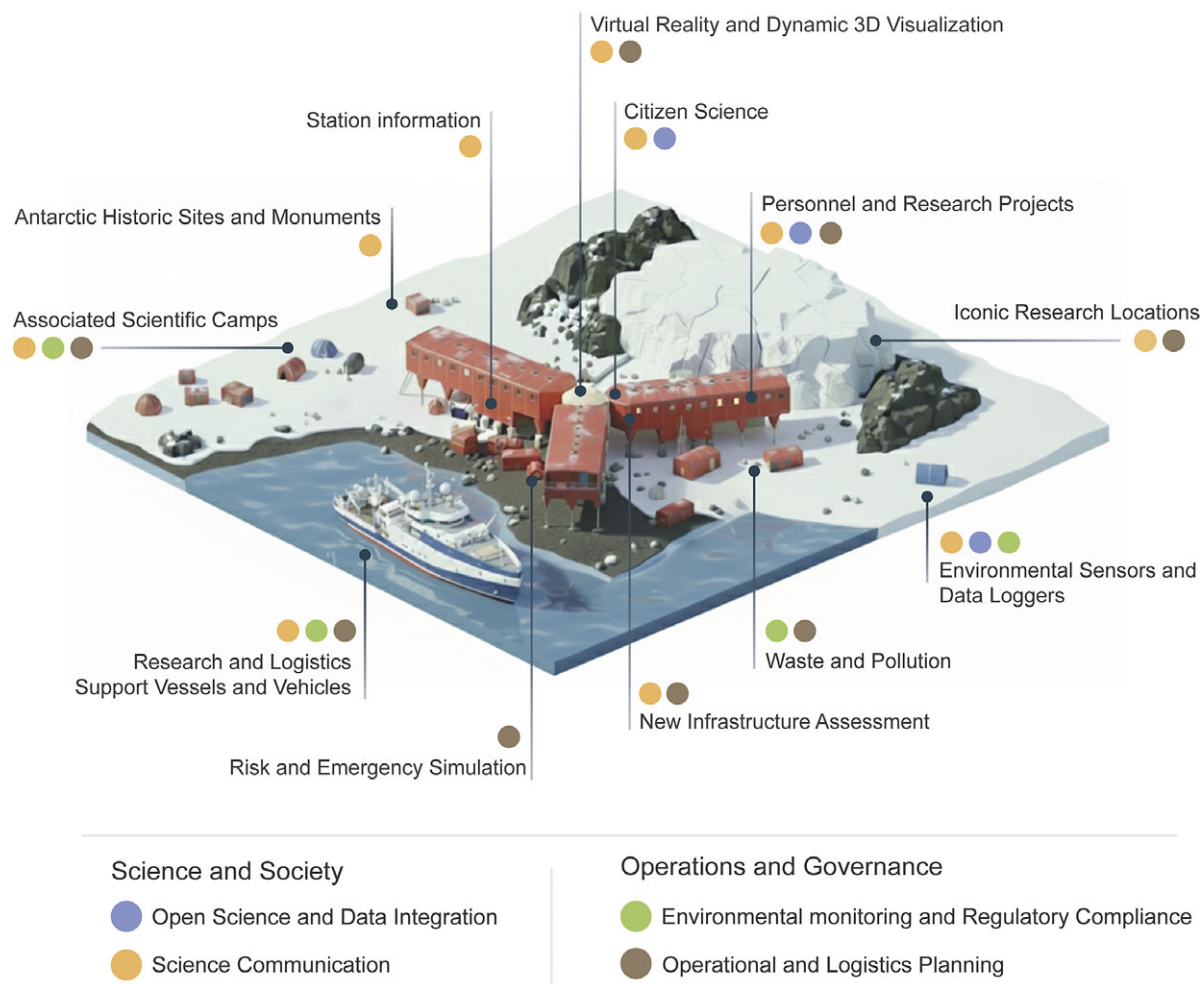


Figure 1. Schematic overview of a digital twin framework applied to an Antarctic research station. The figure illustrates the integration of environmental sensors, modular station buildings, support infrastructure, logistics and research activities and surrounding terrain within a spatial digital twin. Their main applications are summarized into four colour-coded categories: open science and data integration (purple), scientific communication (light orange), environmental monitoring and regulatory compliance (green) and operational and logistics planning (brown). The illustration is based on the Juan Carlos I Antarctic research station (Livingston Island) and the Spanish research/logistical vessel *Odón de Buen* (IEO-CSIC).

be complemented by virtual tours of the interiors of these facilities (Fig. 2; Hu & Assaad 2024). They can also include 3D visualizations of historical or scientific areas of interest, such as Antarctic Historic Sites and Monuments (HSM), or seasonal research camps located near the primary research facility (e.g. the Spanish international Byers Camp on Livingston Island). Moreover, these 3D models enable the representation of historical scenarios and future projections, illustrating the addition of new buildings or the removal of outdated ones, laboratory expansions and updates to renewable energy systems (Boccaletti *et al.* 2014, Davis 2017), as well as changes in operational capacity over the years (Fig. 3a).

Along the same lines, SDTs can integrate or link to 360° virtual reality (VR) imagery and video platforms (Caruso & Schembra 2024). Examples of existing VR experiences in polar regions include immersive reconstructions of research station surroundings (e.g. VR visit to Spanish Gabriel de Castilla research station on Deception Island; <https://ejercito.defensa.gob.es/videos/2021/antartida/web/index.htm>), as well as simulations of environments of scientific interest such as breeding colonies, marine habitats, archaeological or palaeontological sites, volcanic areas and glacier

fronts (e.g. the POLARCSIC 360VR project; <https://www.youtube.com/watch?si=R6B4iUaAmEY8vf9&v=TenTISKoOkw&feature=youtu.be>; Fig. 2). By showcasing these unique and often inaccessible locations, VR experiences represent a powerful tool for outreach and education and may also contribute to improved awareness and support for systematic conservation planning surrounding research stations (Brooks *et al.* 2024, Salinas *et al.* 2024).

Regardless of the context, accessibility must be a central pillar of SDT-based outreach. SDTs should be designed to provide multi-lingual interfaces, offering content not only in major international languages (e.g. official languages of the Antarctic Treaty System) but also in national or co-official languages of the stations' nationalities, thereby broadening access for diverse audiences (e.g. Catalan, Basque or Galician in the case of Spain; Figs 2 & 3a). For people with disabilities, SDTs can incorporate assistive technologies such as screen reader compatibility, audio descriptions, adjustable font sizes and high-contrast modes. These features help ensure that scientific outreach is inclusive, allowing anyone, regardless of linguistic or physical limitations, to access and engage with Antarctic research.

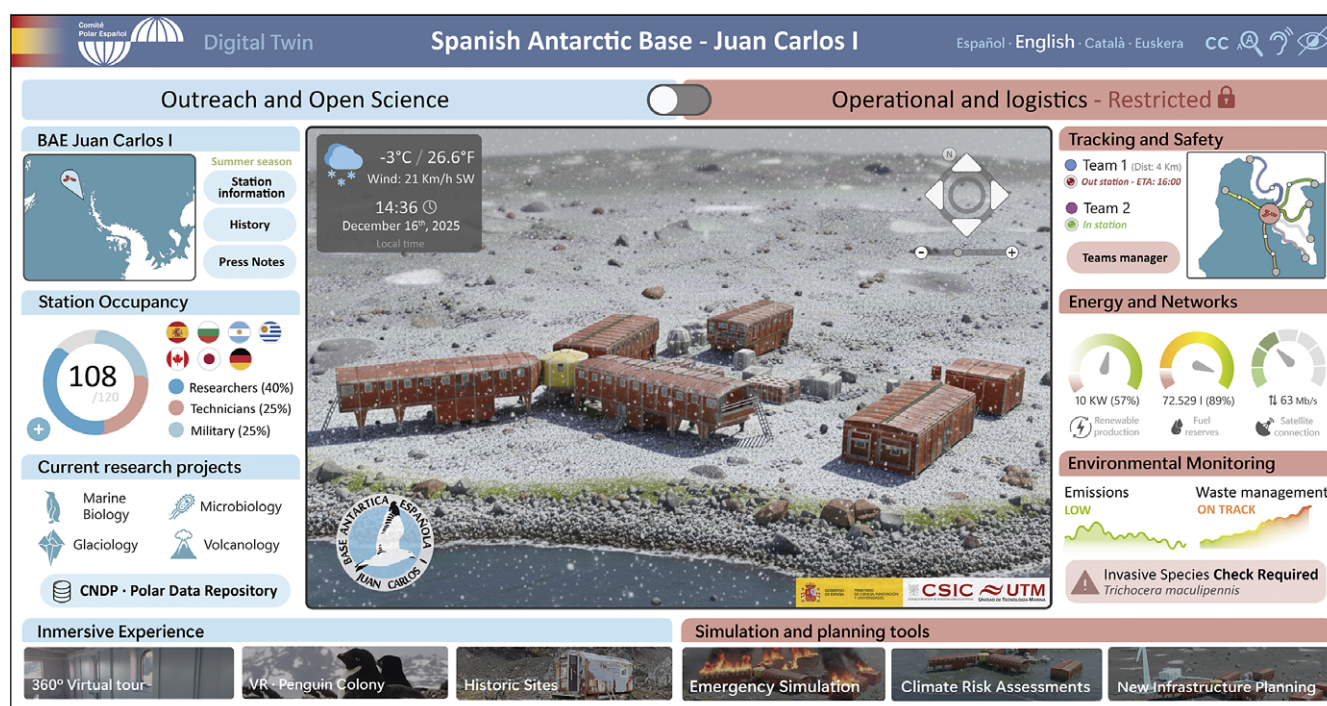


Figure 2. Conceptual dashboard design applied to an Antarctic research station digital twin. The interface distinguishes between public-facing outreach (blue tones) and restricted operational (red tones) components. The public interface provides information on station location, history, outreach materials, current occupancy, active research projects and immersive experiences (e.g. virtual reality), incorporating accessibility and multilingual tools. The operational layer features real-time monitoring of environmental parameters, energy resources and logistics, alongside simulation tools for emergency and infrastructure planning. Both components are unified by an interactive 3D model of the station's infrastructure. The conceptual design is based on the Spanish Juan Carlos I Antarctic research station as a reference.

Spatial digital twins as hubs for open science

In the era of open access and scientific e-infrastructures, the availability and sharing of research data have become fundamental pillars in the knowledge generation, credibility and accessibility of science (Powers & Hampton 2019, Ivimey-Cook *et al.* 2025). SDTs can play a strategic role as central hubs, connecting scientific outputs, open data repositories, application programming interfaces (APIs) and publications, facilitating an integrated and reusable ecosystem (Lehner & Dorffner 2020). For instance, the Spanish Polar Data Centre (Centro Nacional de Datos Polares; cndp.utm.csic.es) exemplifies a repository that can be integrated into a research station SDT (Fig. 2). By linking spatial context, metadata and real-time or historical datasets, SDTs allow users to explore data in an intuitive manner, enhancing data discoverability and reuse. Thus, SDTs contribute to turning data into usable knowledge and promote international collaboration in Antarctic research (Jang *et al.* 2020, Vila *et al.* 2025).

Beyond formal research outputs, SDTs can also enhance the visibility of citizen science initiatives. They can do this by centralizing access to information and linking together projects that operate at the interface between outreach and open science. Initiatives such as Seal (www.zooniverse.org/projects/cmfoley/seal-watch) or Penguin Watch (www.zooniverse.org/projects/penguintom79/penguin-watch; Jones *et al.* 2018), where volunteers identify pinnipeds and adult penguins, chicks and eggs from camera-trap imagery, respectively, illustrate the scientific value of citizen-contributed data in the context of remote polar areas. Integrating such projects within SDTs would not only connect them with institutional research efforts but also provide a richer spatial, environmental and operational context than that which is typically available on standalone project websites, thereby enhancing the

transparency, interpretability and reusability of citizen science contributions.

Operational and logistical planning

Conceptually, SDTs can be organized into a public-facing layer for outreach and communication and a restricted operational layer with role-based access dedicated to logistics, monitoring and safety management (Figs 2 & 3; Crespi *et al.* 2023). By integrating real-time positioning systems (e.g. Automatic Identification System (AIS) and Automatic Dependent Surveillance - Broadcast (ADS-B)) they enable continuous monitoring of supply routes for ships and other tracked vehicles (aircraft or terrestrial vehicles). As a novel operational tool, SDTs can dynamically update the location of scientific and technical personnel deployed in the field using satellite-based global communication and positioning systems in remote areas (e.g. Iridium), thereby enhancing personnel safety, situational awareness and overall logistical efficiency while improving human occupancy models for the research station (Fig. 2). In addition to their use in outreach, the 3D visualizations and models described earlier provide practical tools for logistical planning and resource optimization within a single integrated virtual environment. Inspired by their application in smart cities, SDTs can also be used to assess climatic and environmental risks affecting infrastructure (e.g. sea-level rise or changes in ice cover), as well as to simulate emergency and evacuation scenarios, such as fires or tsunamis, ultimately improving operational safety (Fig. 2). Furthermore, they enable the assessment of the impact of new infrastructure at established stations (e.g. landing sites, energy installations, buildings or laboratories; Fig. 3b; Boccaletti *et al.* 2014), and they can even support the planning of entirely new stations (Xiaoping *et al.* 2014).



Figure 3. Conceptual detailed visualization of two interactive components of the Antarctic research station digital twin. **a.** History section illustrating the establishment of the research station, together with an interactive virtual model representing some of the former infrastructure present at that time. In the proposed concept, this visualization is complemented by textual resources and historical images to contextualize past periods. The interface is designed to be interactive, allowing users to explore and visualize changes in the station's infrastructure over time. **b.** Conceptual planning tool for new infrastructure within the station's digital twin, illustrated through the example of implementing a new wind turbine. This component showcases how the information provided by the digital twin can support infrastructure planning and decision-making processes. Both conceptual designs are based on the Spanish Juan Carlos I Antarctic research station.

Environmental monitoring and regulatory compliance

Scientific activity and human presence in Antarctica inevitably have environmental impacts and footprints arising from the operation of research stations (Crossin *et al.* 2020). In this context,

SDTs enable the integration of observational data with different models, facilitating more accurate, near-real-time environmental monitoring, as has been demonstrated in other contexts (Lehner & Dorffner 2020). By doing so, SDTs can help evaluate the environmental effects of human activities on the continent

(including emissions, fuel consumption and other operational impacts) while supporting the preparation of environmental reports in accordance with the Antarctic Treaty Madrid Protocol (Fig. 2; The Secretariat of the Antarctic Treaty 1991, Arsecularatne *et al.* 2024). Furthermore, they facilitate the simulation and tracking of pollutant spills, as well as waste management from generation to final disposal, in a traceable and quantifiable manner (Hughes 2003, Emnet *et al.* 2015). Another key aspect that can be integrated into these systems is the detection and monitoring of invasive species, such as the Antarctic mosquito, whose presence has been primarily recorded in research stations (Volonterio *et al.* 2013, Lee *et al.* 2024). Collectively, SDTs optimize environmental management, strengthen regulatory compliance and facilitate rapid, cross-station alerts for emerging risks, fostering transparent and informed research community compliance in extreme environments.

Challenges and limitations

Despite their potential and multiple applications, the implementation of SDTs in Antarctic environments faces several challenges, many of which are inherent to their remoteness and directly affect real-time monitoring applications. Limited connectivity, high energy demands, physical maintenance constraints and the risk of sensor disconnection during the winter, when many bases and much infrastructure remain closed, significantly impact system reliability. Furthermore, the diversity of the polar programmes involved, the research stations and the technological systems and architectures could complicate the establishment of a single and standardized SDT model. The Spanish Juan Carlos I Antarctic research station has been used here as an illustrative example of the broader applicability of this approach due to its historical relevance, operational capacity and recent modernization, among other characteristics. The information presented is intentionally generic and does not aim to accurately represent the current state of the station but rather to highlight the broader applicability of the proposed SDT approach to other Antarctic research stations. Thus, standardization of data formats, exchange protocols and monitoring methodologies is crucial to ensuring interoperability among different actors. In this context, the data hosted within a SDT could involve the management of sensitive information, requiring robust protocols against security vulnerabilities (Evangeline 2025), thereby guaranteeing transparency without compromising the outreach aims and operational security.

In addition, a primary barrier to SDT implementation is the acquisition bottleneck, particularly the reliance on manual inputs for operational workflows. Although automated sensor networks effectively capture environmental data, the human-in-the-loop requirement for logistical and activity-based records remains a significant hurdle. While the proposed framework is well-suited for small- to medium-sized research stations with defined operational scopes, scaling this system to larger, more complex infrastructure presents substantial challenges. To remain viable, SDT development must prioritize high-level automation by leveraging existing digital logs and interoperable software to minimize the administrative burden on station staff. In small- to medium-sized research stations, every staff member is a critical resource; therefore, relying on manual data entry not only is inefficient but also creates a direct trade-off with scientific personnel capacity. To bridge this gap, the system should pivot from manual human-input workflows towards autonomous data acquisition. A compelling example of this shift is the automation

of human occupancy modelling highlighted earlier - a novel application proposed for SDTs in terms of both public outreach and the management of scientific activities (see Fig. 2). By integrating diverse movement tracking technologies, such as indoor LiDAR-based positioning (Karki *et al.* 2024), localized proximity sensors (e.g. radio frequency identification; RFID) for internal zones and satellite-based geolocation for field operations, the system can autonomously monitor human occupancy. This approach continuously tracks personnel across both interior facilities and remote field sites, replacing manual reporting with real-time data. This significantly improves the spatial mapping of human activity, which is essential for ensuring the protection of critical areas and the biodiversity surrounding research stations (Salinas *et al.* 2024).

Nonetheless, a specific cost-benefit analysis on the research station scale is essential, weighing initial outlays against long-term gains in safety, logistical optimization and scientific transference. From an institutional perspective, this investment should not be viewed as an isolated expenditure but as a strategy to reduce administrative overheads and enhance operational resilience. By prioritizing a modular architecture, national polar programmes can ensure incremental value gains at every stage, effectively mitigating the risk of prohibitive upfront costs. Crucially, this financial commitment must be contextualized within the broader scale of polar operations: the cost of developing and maintaining a digital twin represents a negligible fraction of outgoings compared to the immense capital required for annual station maintenance, summer campaigns and vessel and/or aircraft logistics. Therefore, SDT funding should not be drawn from short-term, competitive research grants; rather, its long-term viability must be anchored directly within the core structural and operational maintenance budgets of national polar programmes or research institutions. As recently highlighted by Morandini & Soutullo (2026), this shift is critical: current funding mechanisms often prioritize project novelty and visibility over institutional responsibility and long-term continuity, a structural weakness that compromises the governance capacity of Antarctic science. By integrating SDTs into these institutional budgets, akin to the sustained support provided for physical infrastructure and field campaigns, these platforms are elevated from temporary project lifecycles to permanent management assets.

Ultimately, the effectiveness of a SDT depends less on technical ingenuity than on this sustained institutional commitment. Beyond technical and financial hurdles, other systemic challenges remain: the risk of algorithmic bias in automated decision-making (Sabri *et al.* 2025), the socio-ethical implications of constant personnel monitoring for base dynamics (Mureddu *et al.* 2025, Cantisani *et al.* 2026) and the administrative complexities of data governance and institutional stewardship among national programmes in the current shifting Antarctic landscape (Haward & Jackson 2023, Dodds 2024a, Arpi 2026). Thus, the transition to a comprehensive SDT is not merely a technical upgrade but an institutional evolution that requires strategic resource alignment. Viewing these inevitable obstacles as evolutionary stages rather than barriers is essential to ensuring that the next generation of Antarctic digital infrastructure remains resilient, accessible and adaptive to the shifting demands of polar science.

Roadmap for spatial digital twin implementation

The deployment of a station SDT should not be viewed as a single process but as an iterative evolution from a static visual asset

to a dynamic operational engine. The following steps should be considered:

- 1) A potential starting point may be the development of a 3D digital model (BIM/DSM (Digital Surface Model)) integrated with drone-based orthophotos of the research station's surroundings (e.g. Román *et al.* 2024). At this stage, a web-based SDT functions primarily as an immersive window for public outreach and scientific communication (station and research activities locations, dashboards and hubs for scientific project data-sharing repositories). By visualizing existing open-access environmental feeds alongside the physical infrastructure, the system gains immediate value with minimal investment. This establishes a baseline for future data integration and functions as a standalone tool for the monitoring and management of station activities, including tracking projects under the Antarctic Treaty System (Salinas *et al.* 2024).
- 2) Having created and stabilized this first 'twin', the process transitions into an active management hub by layering restricted, role-based operational data. This involves moving from static visualization to real-time monitoring, integrating logistical flows (e.g. human occupancy, vessel/aircraft tracking), infrastructure maintenance logs and sensor networks. This is where the SDT proves its utility, improving the science dissemination features (with near-real-time data) and transitioning to a decision-support system that optimizes station safety and logistics.
- 3) In the long term, the system advances towards predictive maintenance and risk assessment (Fig. 3b) and establishes an interoperable data-sharing agreement that aligns with the Antarctic Treaty's principles; individual station twins can eventually converge into a regional or common sovereignty. While this modular approach is designed for individual stations, it inherently lays the groundwork for regional synergy. As these twins mature, they could be integrated initially among stations sharing the same national programme (e.g. Profesor Julio Escudero (administered by Instituto Antártico Chileno, INACH) and Presidente Eduardo Frei Montalva (administered by Fuerza Aérea Chilena, FACH), both Chilean stations at King George/25 de Mayo Island, Antarctic Peninsula), eventually extending to neighbouring international infrastructure that shares logistical chains. Although such 'regional twins' represent a long-term aspiration, the standardization of digital twin protocols proposed here provides the basis for this collaborative digital ecosystem.

Final considerations

At present, (spatial) digital twins are emerging as novel and transformative tools for management, research and outreach (Crespi *et al.* 2023), and they have demonstrated great potential for their application in Antarctic logistical and research activity and their wider dissemination. The scientific ecosystem that governs the human presence in Antarctica provides an ideal setting to maintain and feed SDTs with well-defined aims. This perspective paper illustrates and highlights several potential applications and uses of SDTs. Crucially, the scope of these platforms should synergize with existing digital initiatives for research and logistical vessels (also illustrated in Fig. 1; e.g. AI and Digital Twinning for Decarbonisation (www.bas.ac.uk/blogpost/seeing-double-digital-twins-and-net-zero/)) for reducing carbon emissions from the polar ship RRS *Sir David Attenborough*).

By linking these mobile assets into the SDT ecosystem, we can capture a more holistic view of Antarctic logistics and scientific operations, effectively bridging the gap between offshore activity and land-based research infrastructure. As these systems evolve, their implementation could progressively expand, enriching their utility across multiple domains and serving a wide range of stakeholders. Their ability to integrate real-time data, model infrastructure, monitor logistical operations and assess environmental risks positions SDTs as essential platforms for decision-making in remote regions.

Although this framework serves as a proof-of-concept for an individual base, there is potential for it to evolve into a regional-scale system. The Antarctic Peninsula, characterized by a high density of international programmes (COMNAP 2017) and shared logistics, represents the logical next step for such a development. Transitioning from a station-specific to a regional model will require addressing significant technical and geopolitical challenges, ranging from international data interoperability to institutional frameworks under the Antarctic Treaty System. Consequently, this work serves as a foundational step: by establishing standardized protocols at a station level, we provide the necessary building blocks for future collaborative, continent-wide digital infrastructure.

Beyond operational management, SDTs also enhance science communication and outreach. As noted by the late Antarctic researcher Andrés Barbosa, 'The polar regions are very attractive, and we must take advantage of this circumstance to disseminate them' (García-Roa 2023). SDT models enable transparent and accessible public communication, bringing Antarctic science closer to the public while promoting education, outreach (Xavier *et al.* 2024, Vila *et al.* 2025), international cooperation and networking within the scientific community (Kim & Jung 2016, Ríos-Silva *et al.* 2025). Much of the information that is available for some research stations and refuges is currently documented in a fragmented manner across multiple websites, blogs and other online sources. SDTs provide the context needed to integrate these disparate data sources, improving accessibility, usability and overall understanding of station activities. Although, as mentioned, the full implementation of digital twins may face technological and logistical limitations, the development of functional prototypes at selected research stations would represent a crucial first step towards the digitalization of polar research infrastructure. Such efforts could contribute to more coordinated, data-rich science, which is crucial in light of the emerging conservation challenges in Antarctica that demand innovative tools and collaborative approaches from fragmented observation into coordinated foresight (Morandini & Soutullo 2026). They could also help to usher in a new era of open, collaborative and data-driven research in one of the planet's most fragile and remote ecosystems (Carter *et al.* 2025).

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Open research. Conceptual virtual models illustrated in the figures were developed using the generative AI model Gemini 2.5 Flash Image (Nano Banana, Google), based on manuscript author- and supervisor-defined

prompts and informed by open-access pictures as well as images taken directly by the author in different Antarctic campaigns. These original images were subsequently modified and integrated into the figures using Adobe Illustrator.

Competing interests. The author declares none.

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